NIELIT Virtual Academy

COURSE PROSPECTUS

Name of the Course: Online Certificate Course in System Verilog and UVM using Synopsys tool
Course Code: VL11
Mode of Conduction: Online
Starting Date: 15th May 2025
Last date of registration: 12th May 2025
Duration: 30 Hours/2 weeks

Objective of the Course

The primary objective of a Certificate Course in Design Verification is to equip students with the necessary skills and knowledge to ensure the functional correctness of complex integrated circuits (ICs) or System-on-Chips (SoCs). Through hands-on experience with Synopsys tools like VCS, participants gain practical exposure to industry-standard environments for functional verification. The course also covers constrained-random testbenches, functional coverage models, and assertions, helping learners master the techniques needed for verifying complex designs

Prerequisites

- ✓ Candidate must have latest computer/laptop with preferably 4 GB RAM or higher and Graphics Card (2 GB)
- ✓ Internet connection with good speed (preferably 2Mbps or higher)
- ✓ Basic knowledge in digital design concepts and semiconductor technology
- ✓ Familiarity with simulation tools and debugging techniques is a plus.

Outcome of the Course:

Upon completion of the course, participants can expect to achieve:

- **Demonstrate a strong understanding** of the verification process, methodologies, and tools.
- **Develop and implement effective verification plans** for complex IC designs.
- **Create robust testbenches** using industry-standard verification languages (Verilog, SystemVerilog, UVM).
- **Utilize advanced verification techniques** like formal verification and assertion-based verification.
- **Analyze and debug design issues** efficiently and accurately.
- **Collaborate effectively with design** and other verification teams.
- **Contribute to the overall success** of IC design projects.

Course Fee: **Rs: 1,000/- (inclusive of GST)**

Eligibility: Perusing B.E./B. Tech in ECE/EE/CS/IT/EI or aligned branches

Or

Perusing B.Sc(Electronics/IT)/M.Sc. (Electronic/IT)

or

2-year Diploma after 12th in ECE/EEE/CS/IT or aligned branches

Methodology:

- ✓ Teaching Mode: Online
- ✓ Instructor-led live sessions
- ✓ Online/ Blended lab sessions
- ✓ Content access through LMS
- ✓ Recorded Session Available (after the instructor's led-live session)

Registration Link: http://nva.nielit.gov.in

Contact Details:

- Course coordinator Name: Chandralakha R Pillai
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Course Structure:

Module No	Module Title	Duration (Hours)
1	System Verilog modelling concepts	5
2	System verilog program and operators	5
4	Test benches and functional coverage	5
5	UVM Environment	5
	Total Duration	***20***

Syllabus:

Module 1: System Verilog modelling concepts

- Signification of Verification
- Verification Methodology and approaches
- System Verilog Assertions,introduction to assertion
- SVA Basics ,immediate and concurrent assertions
- Sequence operators-AND,OR,repeat,if else

Module 2: System Verilog program and operators

- Data types, declarations, syntax rules
- Procedural blocks and assignments
- Programming statements
- Operators and operation rules
- Compound data types and packages
- System Verilog interface ports

Module 3: Test benches and functional coverage

- Verification constructs and testbench interfaces
- Verification timing using clocking blocks
- Object Oriented test benches
- Functional coverage
- Synchronization
- Assertions overview

Module 4: UVM environment

- UVM Test bench Architecture
- UVM Class library
- Transaction level modelling
- Transaction level components
- Driver and sequencer
- Mini Project

Examination & Certification:

Theory	Practical	Project		Total
No. of Papers:01	No. of Papers:01	Project	Presentation	
20	40	20	20	100

After successful completion of the course, candidate will get an online certificate with the following Grading Scheme:

Grade	S	A	B	C	D	Participation Certificate
Marks Range (in %)	85 to 100	75 to 84	65 to 74	55 to 64	50 to 54	Below 50